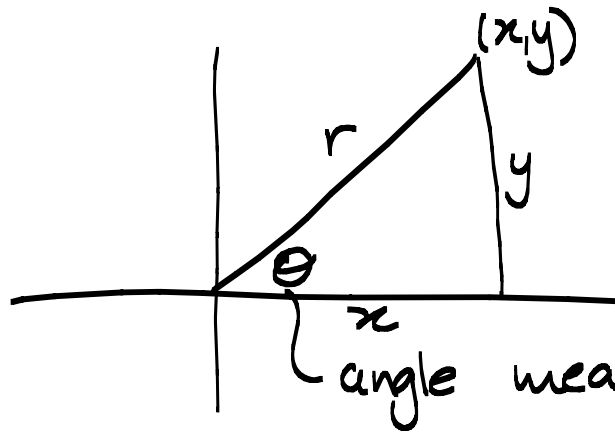


Radian measure & Trig functionsTrigonometry

$$\sin \theta = \frac{y}{r} \quad \left(\frac{O}{H} \right)$$

$$\cos \theta = \frac{x}{r}$$

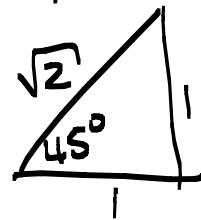
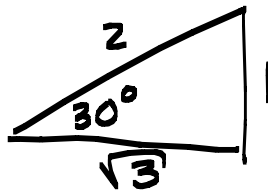
$$\tan \theta = \frac{y}{x}$$

angle measured in degrees.

S	A
T	C

Example:

Special angles:



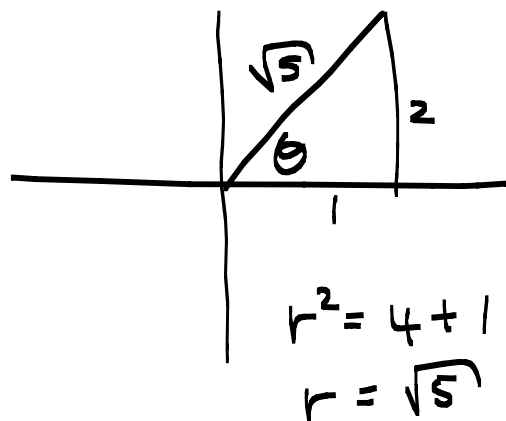
① If $\tan \theta = \frac{2}{1}$ determine $\sin \theta$ &

$$\cos \theta .$$

$$\sin \theta = \frac{2}{\sqrt{5}}$$

$$\cos \theta = \frac{1}{\sqrt{5}}$$

$$\downarrow \frac{1}{\sqrt{5}} \left(\frac{0}{a} \right)$$



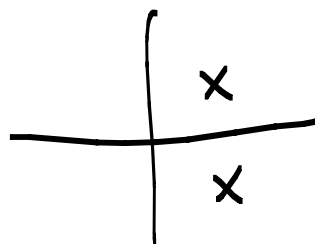
② Solve x if

$$2\cos x - 1 = 0$$

$$, \quad x \in [0^\circ, 360^\circ]$$

$$\Rightarrow \cos x = \frac{1}{2}$$

Ref angle 60°



$$x = 60^\circ \quad \text{or} \quad x = 360^\circ - 60^\circ = 300^\circ.$$

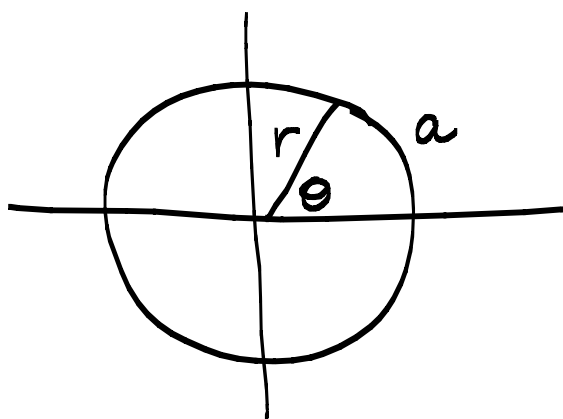
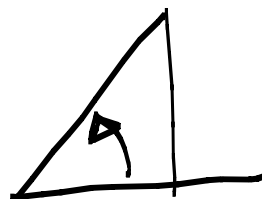
Solve for x :

$$\sin x + x = 2. \quad ?!?$$

$$x = 30^\circ$$

$$\sin 30^\circ + 30^\circ?$$

$$\frac{1}{2} + 30^\circ?$$



$$\theta = \frac{a}{r} \sim \text{metre.}$$

$\sim$ no units. - radians.

$$360^\circ = \frac{2\pi r}{r} = 2\pi$$

$$\Rightarrow \boxed{180^\circ = \pi}$$

$$1^\circ = \frac{\pi}{180} \text{ rad.}$$

$$\frac{180^\circ}{\pi} = 1 \text{ rad.}$$

$$0^\circ = 0$$

$$30^\circ = \frac{\pi}{6}$$

$$60^\circ = \frac{\pi}{3}$$

$$360^\circ = 2\pi$$

$$45^\circ = \frac{\pi}{4}$$

$$90^\circ = \frac{\pi}{2}$$

$$180^\circ = \pi$$

$$720^\circ = 4\pi$$

$$\sin \frac{5\pi}{6} = +\frac{1}{2}$$

$$\tan \frac{2\pi}{3} = -\sqrt{3}$$

$$\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\tan \frac{9\pi}{2} \text{ not defined.}$$

$$\begin{aligned} & \sin\left(\frac{50\pi}{2}\right) \\ &= \sin(25\pi) \\ &= \sin(24\pi + \pi) \end{aligned}$$

$$\begin{aligned} & \cos\left(\frac{3\pi}{4}\right) \\ &= -\frac{1}{\sqrt{2}} \end{aligned}$$

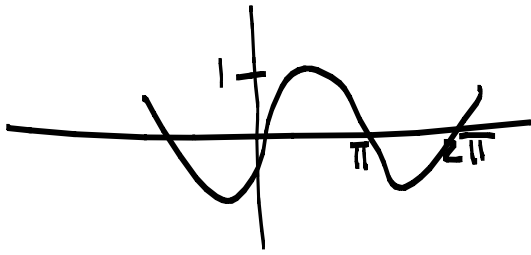
$$= \sin \pi = 0$$

$$\tan\left(\frac{7\pi}{3}\right)$$

$$= + \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}$$

Trigonometric functions:

$$y = \sin x$$



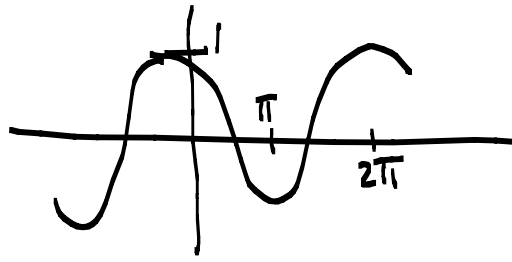
$$-1 \leq \sin x \leq 1$$

$$\rightarrow |\sin x| \leq 1$$

$$x \in \mathbb{R}$$

$$y \in [-1, 1]$$

$$y = \cos x$$



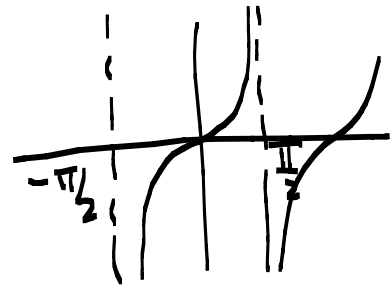
$$-1 \leq \cos x \leq 1$$

$$|\cos x| \leq 1$$

$$x \in \mathbb{R}$$

$$y \in [-1, 1]$$

$$y = \tan x$$



$$x \in \mathbb{R} \setminus \left\{ \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots \right\}$$

$$y \in \mathbb{R}$$

Trig. reciprocals:

$$\frac{1}{\sin \theta} = \operatorname{cosec} \theta \quad (\csc \theta) \quad \left(\frac{r}{y}\right)$$

$$\frac{1}{\cos \theta} = \sec \theta \quad \left(\frac{r}{x}\right)$$

$$\frac{1}{\tan \theta} = \cot \theta \quad \left(\frac{x}{y}\right)$$

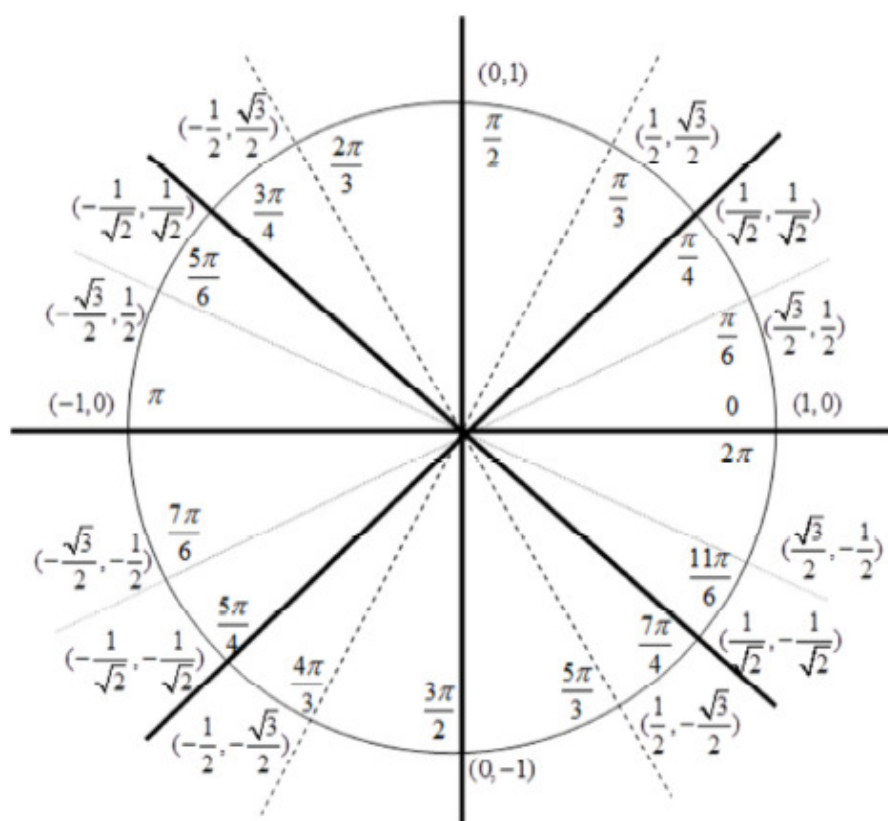
$$\sin \theta \in \left(\frac{y}{r}\right)$$

Q1 Find

$$1, \cot\left(-\frac{2\pi}{3}\right) = +\frac{1}{\sqrt{3}}$$

$$2, \sec\left(-\frac{2\pi}{3}\right) = -2$$

$$\text{OR} \\ \sec\left(-\frac{2\pi}{3}\right) = \sec\left(\frac{2\pi}{3}\right) = -2$$



$$1. \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} \quad 2. \sin \frac{3\pi}{4} = \frac{1}{\sqrt{2}}$$

$$3. \cos \frac{7\pi}{4} = \frac{1}{\sqrt{2}} \quad 4. \sin \left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$5. \cos \frac{5\pi}{6} = -\frac{\sqrt{3}}{2} \quad 6. \tan \left(\frac{2\pi}{3}\right) = -\sqrt{3}$$

$$7. \sin \left(\frac{3\pi}{2}\right) = -1 \quad 8. \sin \left(\frac{11\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

Finish the ones on the worksheet.

Solving trigonometric equations.

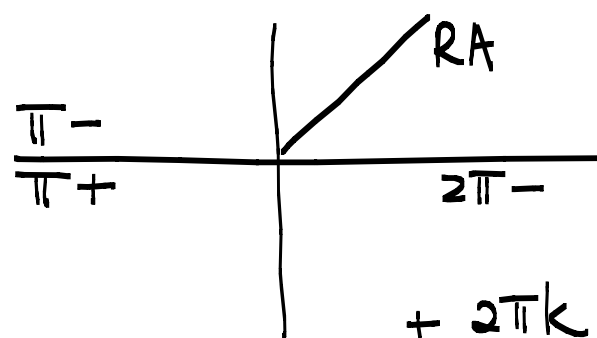
1. Solve for $\theta \in [0, 2\pi]$ if *

$$\sin \theta = -\frac{1}{\sqrt{2}}$$

$$\text{R.A: } \theta = \frac{\pi}{4}$$

$$\Rightarrow \theta = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$$

$$\text{or } \theta = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$$



$$+ 2\pi k, \quad k \in \mathbb{Z}$$

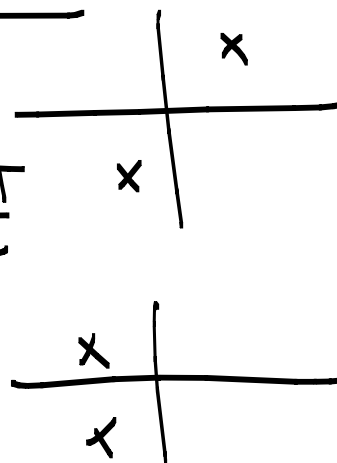
$$2. \tan \alpha = 1$$

$$0 < \alpha < 2\pi$$

$$\text{RA} = \frac{\pi}{4}$$

$$\therefore \alpha = \frac{\pi}{4} \text{ or } \alpha = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$$

$$3. \cos \alpha = -\frac{\sqrt{3}}{2}$$



$$RA: \frac{\pi}{6}$$

$$\Rightarrow x = \pi - \frac{\pi}{6} \quad \text{or} \quad x = \pi + \frac{\pi}{6}$$

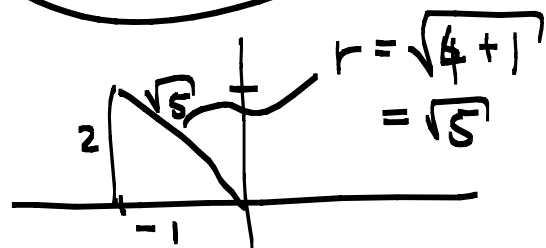
$$= \frac{5\pi}{6} + 2k\pi \quad = \frac{7\pi}{6} + 2k\pi$$

Where $k \in \mathbb{Z}$.

4. If $\tan \theta = -2$ $\left(-\frac{2}{1}\right)$ $\left(\frac{\pi}{2} < \theta < \pi\right)$

find $\cos \theta$.

$$\cos \theta = \frac{-1}{\sqrt{5}}$$



5. $\operatorname{cosec} x = \frac{-2}{\sqrt{3}}$

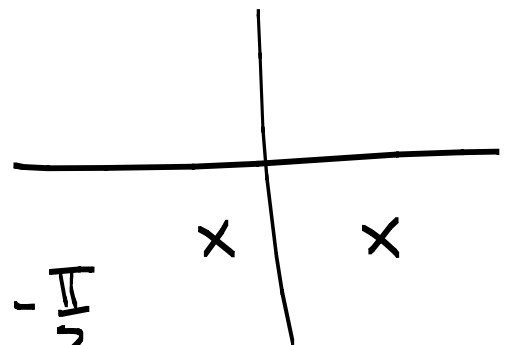
$$\therefore \sin x = -\frac{\sqrt{3}}{2}$$

(Solve for x .)

$$RA: x = \frac{4\pi}{3}$$

$$\Rightarrow x = \pi + \frac{\pi}{3} \quad \text{or} \quad x = 2\pi - \frac{\pi}{3}$$

$$= \frac{4\pi}{3} + 2k\pi \quad x = \frac{5\pi}{3} + 2k\pi$$



6. $\cot 2\alpha = \sqrt{3}$

$$\alpha \in [0, \pi]$$

$$RA \quad 2\alpha = \frac{\pi}{6}$$

$$\tan 2\alpha = \frac{1}{\sqrt{3}}$$

$$\therefore 2\alpha = \frac{\pi}{6} \quad \text{or} \quad 2\alpha = \pi + \frac{\pi}{6}$$

$$\alpha = \frac{\pi}{12} \quad \checkmark \quad 2\alpha = \frac{7\pi}{6}$$

$$7. \frac{1}{2} \sec^2 \theta = 1$$

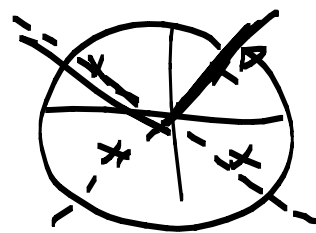
$$\sec^2 \theta = 2$$

$$\Rightarrow \sec \theta = \pm \sqrt{2}$$

$$\theta = \frac{\pi}{4} + 2k\pi$$

$$= \frac{\pi}{4} + \frac{\pi}{2}k, \quad k \in \mathbb{Z}$$

$$\theta = \left(\frac{\pi}{12} \right) \checkmark$$



.... -2, -1, 0, 1, 2, 3, ...

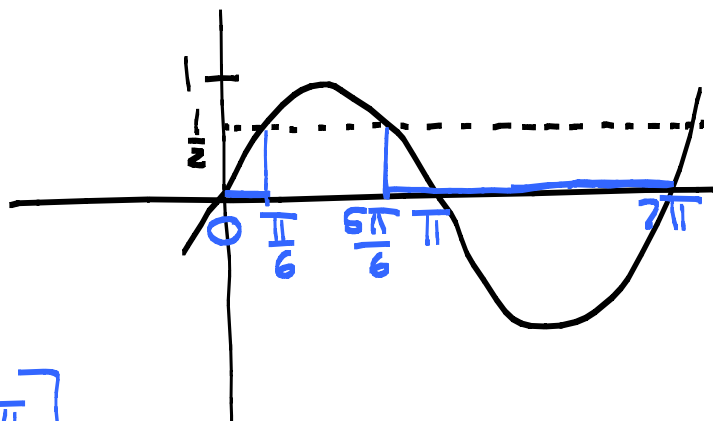
Trig inequality. Find the values
of $x \in [0, 2\pi]$ so that

$$1. \sin x \leq \frac{1}{2}$$

$$\text{Consider } \sin x = \frac{1}{2}$$

$$x = \frac{\pi}{6} \text{ or } x = \frac{5\pi}{6}$$

$$x \in \left[0, \frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}, 2\pi\right]$$



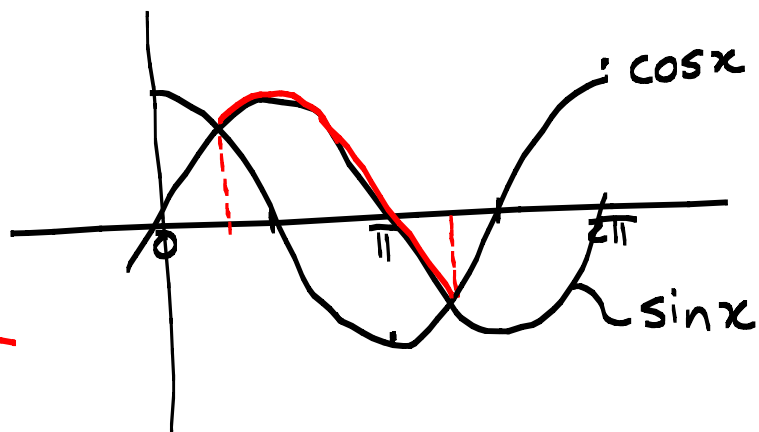
$$2. \sin x > \cos x$$

$$x \in [0, 2\pi]$$

$$\text{Consider } \frac{\sin x}{\cos x} = \frac{\cos x}{\cos x}$$

$$\Rightarrow \tan x = 1$$

$$\therefore x = \frac{\pi}{4} \text{ or } x = \frac{5\pi}{4}$$



$$\sin x > \cos x \text{ for } x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$$

SKETCH!!!

3. Solve

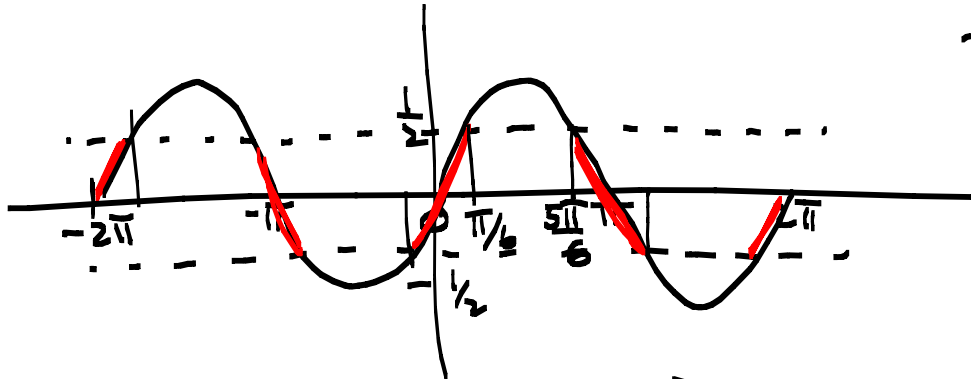
$$|\sin x| \leq \frac{1}{2}, \quad x \in [-2\pi, 2\pi]$$

$$\Rightarrow -\frac{1}{2} \leq \sin x \leq \frac{1}{2}$$

Consider

$$\sin x = \pm \frac{1}{2}$$

$$x = \frac{\pi}{6}, \dots$$



$$x \in [-2\pi, -\frac{11\pi}{6}] \cup [-\frac{7\pi}{6}, -\frac{5\pi}{6}] \cup [-\frac{\pi}{6}, \frac{\pi}{6}] \cup$$
$$[\frac{5\pi}{6}, \frac{7\pi}{6}] \cup [\frac{11\pi}{6}, 2\pi]$$

Worked exemplar on ClickUP under my name!

Solve.

$$x \in [0, 2\pi]$$

$$\textcircled{1} \quad 2 + \cos 2x = 3 \cos x$$

$$2 + \cos^2 x - \sin^2 x = 3 \cos x$$

$$2 + \cos^2 x - (1 - \cos^2 x) = 3 \cos x$$

$$2 \cos^2 x - 3 \cos x + 1 = 0$$

$$(2 \cos x - 1)(\cos x - 1) = 0$$

$$\Rightarrow \cos x = \frac{1}{2} \quad \text{or} \quad \cos x = 1$$

$$x = \frac{\pi}{3}, \left(2\pi - \frac{\pi}{3}\right)$$

$$\cos x = 1$$

$$x = 0, 2\pi$$

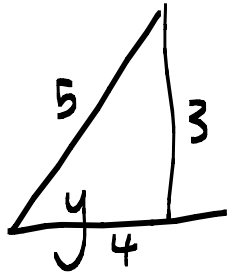
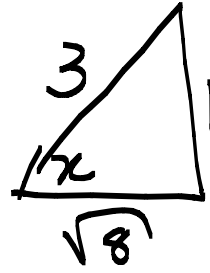
② If $\sin x = \frac{1}{3}$ and $\sec y = \frac{5}{4}$
 and $x, y \in \left[0, \frac{\pi}{2}\right]$. $\cos y = \frac{4}{5}$

Evaluate $\sin(\overset{\downarrow}{x} + \overset{\downarrow}{y})$

$$= \sin x \cos y + \cos x \sin y$$

$$= \left(\frac{1}{3}\right)\left(\frac{4}{5}\right) + \left(\frac{\sqrt{8}}{3}\right)\left(\frac{3}{5}\right)$$

$$= \frac{4}{15} + \frac{\sqrt{8}}{5} = \frac{4 + 3\sqrt{8}}{15}$$



Important! Trig identities.

7

$$\sin^2\theta + \cos^2\theta = 1$$

8

$$\tan^2\theta + 1 = \sec^2\theta$$

15a

$$\sin 2x = 2 \sin x \cos x$$

15b

$$\cos 2x = \cos^2x - \sin^2x$$

17a

$$\cos^2x = \frac{1 + \cos 2x}{2}$$

17b

$$\sin^2x = \frac{1 - \cos 2x}{2}$$